ANOVA: Single Factor

Complete Step-by-Step Calculation for Beginners

Dataset: Group A, Group B and Group C. This PDF shows the calculation from raw data to F-value, F-critical, P-value and final decision.

Group A	Group B	Group C
72	80	68
75	82	70
78	85	72
70	79	65
74	84	71
77	81	69
73	83	73
76	86	67
71	80	70
75	82	66

Step 1 — Count the observations

Each group has 10 observations.

```
nA = 10    nB = 10    nC = 10
```

Total observations: N = 10 + 10 + 10 = **30**

Number of groups: k = **3**

Step 2 — Calculate the Sum

Group A: 72 + 75 + 78 + 70 + 74 + 77 + 73 + 76 + 71 + 75 = **741**

Group B: 80 + 82 + 85 + 79 + 84 + 81 + 83 + 86 + 80 + 82 = **822**

Group C: 68 + 70 + 72 + 65 + 71 + 69 + 73 + 67 + 70 + 66 = **691**

Step 3 — Calculate the Mean

```
Formula: Mean = Sum / Count
```

Group A: 741 / 10 = **74.1**

Group B: 822 / 10 = **82.2**

Group C: 691 / 10 = **69.1**

Step 4 — Calculate the Grand Mean

Total sum = 741 + 822 + 691 = **2254**

Grand Mean = 2254 / 30 = **75.1333**

Step 5 — Calculate Variance / Squared Deviations

For beginner practice, first calculate X – Mean, then square it: $(X - \text{Mean})^2$.

Group	Mean	Sum of $(X - \text{Mean})^2$	Sample Variance
A	74.1	61.9	61.9 / 9 = 6.8778
B	82.2	47.6	47.6 / 9 = 5.2889
C	69.1	59.9	59.9 / 9 = 6.6556

```
Sample variance formula: s² = Σ(X − Mean)² / (n − 1)
```

Important: In ANOVA, the SS values are used to calculate the Mean Squares (MS), which act as the variance estimates for the F-test.

Step 6 — Calculate Within-Group Sum of Squares

```
SSWithin = SSA + SSB + SSC
```

= 61.9 + 47.6 + 59.9 = **169.4**

Step 7 — Calculate Between-Group Sum of Squares

```
Formula: SSBetween = Σ n(Mean − Grand Mean)²
```

Group A: $10(74.1 - 75.1333)^2$ = **10.6778**

Group B: $10(82.2 - 75.1333)^2$ = **499.3778**

Group C: $10(69.1 - 75.1333)^2$ = **364.0111**

SSBetween = 10.6778 + 499.3778 + 364.0111 = **874.0667**

Step 8 — Calculate Total Sum of Squares

```
SSTotal = SSBetween + SSWithin
```

= 874.0667 + 169.4 = **1043.4667**

Step 9 — Calculate Degrees of Freedom

Between: dfB = k – 1 = 3 – 1 = **2**

Within: dfW = N – k = 30 – 3 = **27**

Total: dfT = N – 1 = 30 – 1 = **29**

Step 10 — Calculate Mean Square (MS)

MSBetween = SSBetween / dfBetween = 874.0667 / 2 = **437.0333**

MSWithin = SSWithin / dfWithin = 169.4 / 27 = **6.2741**

These MS values are the variance estimates used by ANOVA.

Step 11 — Calculate F-statistic

```
F = MSBetween / MSWithin
```

F = 437.0333 / 6.2741 = **69.657**

Step 12 — Calculate F-Critical

Use significance level α = 0.05, with df1 = 2 and df2 = 27.

```
F-critical = F■¹(0.95, 2, 27) = 3.3541
```

```
Excel: =F.INV.RT(0.05,2,27)
```

Step 13 — Calculate P-value

P-value = P(F ≥ 69.657) with df1 = 2 and df2 = 27.

```
P-value ≈ 2.19 × 10■¹¹ = 0.0000000000219
```

```
Excel: =F.DIST.RT(69.657,2,27)
```

Step 14 — Final ANOVA Table

Source	SS	df	MS	F
Between Groups	874.067	2	437.033	69.657
Within Groups	169.400	27	6.274	
Total	1043.467	29		

Step 15 — Decision

F calculated = 69.657 and F critical = 3.3541.

69.657 > 3.3541 → **Reject H■**

P-value = 0.0000000000219 and α = 0.05.

P-value < 0.05 → **Reject H■**

Final Conclusion

There is a statistically significant difference among the three group means. The group means are A = 74.1, B = 82.2 and C = 69.1.

Easy Memory Trick

Data → Count → Sum → Mean → Grand Mean → Variance/SS → SS Between & Within → df → MS → F → F-critical → P-value → Decision